# Laboratory 7: Simple Processor [8 bit]

## 1 INTRODUCTION

In this lab you will be creating an 8 bit microprocessor with basic addition, subtraction, multiply, and divide operations. You are encouraged to use aliases for the instruction op-codes. The technical requirements for this lab are listed below.

1. **Shall** employ an 'execute' push button.
2. On an 'execute' button push, **shall** fetch and execute an instruction from an inferred block ROM.
    a. The opcode shall be stored in the upper bits of the instruction.
        i. add, subtract, divide, multiply
    b. The second input to the processor ALU shall come from the lower 8 bits of the instruction.
3. The first input into the processor **shall** come from the latest result stored in memory. Coming out of reset, the latest result stored in the 00 register **shall** be zero.
4. **Shall** implement an 8 bit wide memory that contains the ALU results. Could create a 4x8 bit memory as shown below. The number or rows in your memory dictates how many results can be stored.

| address | reg description |
|---|---|
| 00 | Resets to zero |
| 01 | First result |
| 10 | Second result |
| 11 | Third result |

5. The present result **shall** be displayed on 3 seven segment LEDs in decimal form.
6. The result **shall** only be updated upon an 'execute' button push.
7. Processor ALU inputs **shall** both be 8 bit and the processor output shall be 8 bit.
8. **Shall** operate with unsigned base 256 numbers. Ex. 0 - 1 = 255
9. All inputs **shall** be synchronized with the 50 MHz clock.
10. **Shall** display the present state via LEDs.

## 2 Pre-Lab [Block Diagram]

***You shall create an order of operations list as well as a block diagram with a state transition diagram accurately mapping out your design. Also bring a printout of the block diagram to your lab class for an open discussion.*** Make sure to include proper synchronization, edge detection, and signal names in your diagram as well as a clear notation indicating the bit width of various busses and whether a block is a component or process. A state transition diagram as well as an order of operations list is also required.

## 3 Hardware

Target your design onto the DE1 SoC and receive a signoff.

## 4 In-Memory Editor Demonstration

Modify an instruction in real-time and receive a signoff.

## 5 Lab Upload => Required

In order to get full credit for the lab, please do the following.

- ☐ Delete the **work** folder that modelsim generates [if you did a simulation] and the **project** and **output_files** folders in the hw folder.
- ☐ Zip up the high level lab folder
- ☐ Post the .zip file to the below link

https://u.pcloud.com/#/puplink?code=2jlXZug9i7JXqPTujPyXqvCjGNFrBxNyy

## 6 Signoffs- Lab 7 Name:_____

| Category | Initials | Date | Points |
|---|---|---|---|
| Block Diagram | | | /20 |
| Demonstration | | | /60 |
| In-Memory Editor Demonstration | | | /20 |
| Final Grade | | | /100 |